

Integrated Simulation and Optimization for Wildfire Containment

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Wildfire containment is an important but challenging task. The ability to predict fire spread behavior, optimize a plan for firefighting resource dispatch and evaluate such a plan using several firefighting tactics is essential for supporting decision making for containing wildfires. In this article, we present an integrated framework for wildfire spread simulation, firefighting resource optimization and wildfire suppression simulation. We present a stochastic mixed-integer programming model for initial attack to generate firefighting resource dispatch plans using as input fire spread scenario results from a standard wildfire behavior simulator. A new agent-based discrete event simulation model for fire suppression is used to simulate fire suppression based on dispatch plans from the stochastic optimization model, and in turn provides feedback to the optimization model for revising the dispatch plans if necessary. We report on several experimental results, which demonstrate that different firefighting tactics can lead to significantly different fire suppression results for a given dispatch plan, and simulation of these tactics can provide valuable information for fire managers in selecting dispatch plans from optimization models before actual implementation in the field.

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1. INTRODUCTION

The occurrence of devastating wildfires in recent years has brought to the forefront the dangers of wildfires to communities and the environment. If not quickly contained, small wildfires can become escaped fires that are too large to contain. Such fires can burn thousands of acres of forest and often result in the tragic loss of human life and costly damage to property. For example, in the largest U.S. wildfire in 2006, the March 2006 East Amarillo Complex wildfire in Texas, 907,245 acres were burned, 80 structures were destroyed, and 12 lives were lost [NIFC 2006]. In addition, it costs about a billion dollars annually in efforts to contain wildfires in the United States alone [NIFC 2006]. According to the National Interagency Fire Center [NIFC 2006], most wildfires are due to human causes as opposed to natural causes. Wildfire managers are responsible for making decisions regarding wildfire containment. In the event of a reported wildfire, they determine what firefighting resources at each base should be dispatched to the wildfire and generate a plan of action and firefighting tactics to employ. Such decision making is difficult due to the short decision-making time, dynamical and uncertain fire behavior, and limited firefighting resources.

The roles of simulation and optimization for wildfire management are widely recognized. However, in the broad body of work on decision support systems for wildfire suppression, the following deficiencies are noticed. The first deficiency is that computer simulations of wildfire (e.g., FARSITE [Finney 1998], BehavePlus [Andrews et al. 2005], and HFire [Morais 2001]) mainly focus on fire behavior simulation. The aspect of fire suppression simulation with realistic tactics has received less attention as compared to that of fire behavior simulation. Among the related work the authors are aware of, FARSITE supports simulations of both ground attack and aerial attack. The effect of direct attack on an active fire-front is simulated using the known fire perimeter positions at two successive time-steps, and an attack crew building line is defined based on the quadrilateral formed by perimeter vertices in the two time steps. Fried and Fried [1996] developed a mathematical model for direct attack and parallel attack, where parallel attack is modeled in the same way as direct attack for a “super” free-burning fire boundary (fbfb) that has a fixed safe distance to the actual fbfb. This model is also used by BehavePlus for fire suppression simulation.

The second deficiency is that optimization for firefighting resource management [Hodgson and Newstead 1978; Maclellan and Martell 1996; Islam and Martell 1998; Donovan and Rideout 2003; Ntamo et al. 2006] and simulation of firefighting tactics [Fried and Fried 1996; Haight and Fried 2007] are generally treated in isolation without an integrated framework. A major effort of fire management is to determine the (optimal) resource deployment plan for initial attack Fried et al. [2006]. The separation of optimization for firefighting resource management and simulation of firefighting tactics often results in suboptimal solutions for wildfire containment. In particular, the optimization algorithm generally determines firefighting resource dispatch plans while leaving out many of the details of firefighting tactics. Thus there is a need to evaluate

such plans before actual implementation in the field. As will be demonstrated later, different firefighting tactics and initial dispatch locations can result in significantly different fire shapes with different fire perimeters and burned areas. With wildfire suppression simulation, fire managers can examine these details by experimenting with different firefighting tactics and initial dispatch locations for a given fire spreading scenario. Some integrated environments developed for supporting wildfire containment include the California Fire Economics Simulator version 2 (CFES2) [Fried et al. 2006]. CFES2 is a stochastic simulation model designed to facilitate quantitative analysis of the potential effects of changes in key components of wildland fire systems, for instance, availability and stationing of resources, dispatch rules, criteria for setting fire dispatch level, and staff schedules. In a recent paper Haight and Fried [2007] evaluate the deployment decisions from a scenario-based standard response model using CFES2. Another example is the Interagency Initial Attack Assessment (IIAA) system [IIAA 2002] that is used to develop budget requests and to identify the most economically efficient level of a given fire management organization.

This article presents an integrated simulation and optimization framework where fire suppression simulation is combined with fire behavior simulation and optimization for firefighting resource management. The framework includes fire behavior simulation to predict fire spread, stochastic optimization to compute optimal dispatch plans of firefighting resources, and fire suppression simulation to evaluate dispatch plans as well as different firefighting tactics. Simulation of fire behavior is carried out using DEVS-FIRE [Ntaimo and Hu 2008], a discrete event system specification (DEVS) [Zeigler et al. 2000]-based simulation environment. DEVS-FIRE computes fire spread parameters such as fire perimeter, area burned, fireline intensity, and flame length. Stochastic programming (SP) [Birge and Louveaux 1997; Ruszczyński and Shapiro 2003] is used to model optimal firefighting resource deployment to bases, and then dispatch to wildfires based on input data from DEVS-FIRE. The SP model generates an optimal mix of firefighting resources to dispatch to a given wildfire. An agent-based simulation model is used to support simulations with several firefighting tactics, including different types of attack, initial deployment locations, and group configurations. Such simulations provide valuable information for fire managers to assess a dispatch plan before actual implementation in the field. Integrating these components that are usually treated in isolation gives rise to a viable approach for decision making under uncertainty for wildfire management. Within this framework, this article focuses on fire suppression simulation and agent-based firefighting models.

The remainder of the article is organized as follows. The next section describes the overall framework within which this work is developed. Section 3 presents an SP model for firefighting resource management, and Section 4 presents an agent-based model for fire suppression simulation. Experimental results are reported in Section 5, and a discussion given in Section 6. The article ends with some concluding remarks in Section 7.

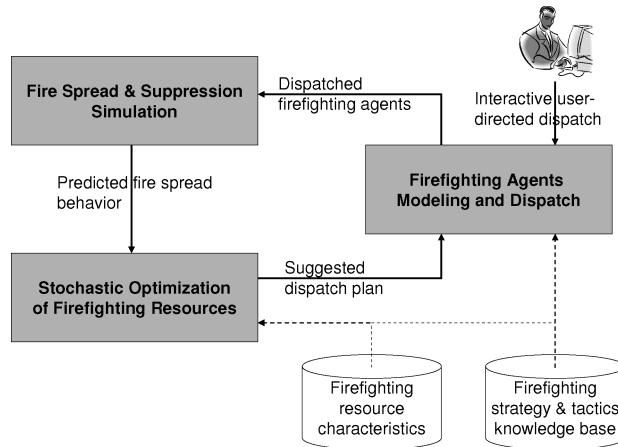


Fig. 1. Integrated simulation and optimization for wildfire containment.

2. AN INTEGRATED SIMULATION AND OPTIMIZATION FRAMEWORK

Simulation of fire suppression relies on the existence of a fire spread model. Meanwhile, a simulation of fire suppression must take the inputs of *what* types of firefighting resources and *when* and to *which* wildfires they should be dispatched. Such dispatch plans of firefighting resources need to be optimized to ensure that a wildfire will be contained as quickly as possible with minimal costs and least damage to the forest area. The dependency among fire spread simulation, fire suppression simulation, and firefighting resource dispatch optimization motivates us to develop an *integrated framework* for supporting wildfire management. Figure 1 shows the three functional components of this framework. Fire spread simulation and fire suppression simulation belong to the same functional component since they are closely related, that is, a fire suppression simulation includes a fire spread simulation. Note that the reverse is not true, as a fire spread simulation can run by itself. The other two components are stochastic optimization of firefighting resources, and firefighting agent modeling and dispatch to the fires.

In the *simulation* component, fire spread simulation allows fire managers to generate predicted fire spreading scenarios, and the fire suppression simulation allows fire managers to experiment with different firefighting tactics. The simulation models of both fire spread and fire suppression are based on DEVS-FIRE. Specifically, the fire spread model is a two-dimensional cellular space model where each cell represents a subarea of the forest. Fire spread is simulated as a propagation process where burning cells ignite their unburned neighbors. The rate of spread of a burning cell is calculated using the Rothermel model [Rothermel 1972] and then decomposed into eight directions corresponding to the eight neighboring cells (Moore neighborhood). Details about the fire spread model can be found in Ntaimo et al. [2004] and are omitted in this article. The fire spread model has been partially validated [Gu et al. 2008]. Simulation

of fire suppression is modeled by agent models, where each group of firefighting resources is modeled as an agent. This results in a hybrid modeling approach that includes both cellular space and agent models. The cellular space model captures the dynamics of wildfire spread while the agent model is responsible for modeling the firefighting actions based on firefighting rules and tactics. This hybrid agent-cellular space modeling approach separates the design concerns of wildfire spread and firefighting making it easy to evolve each one independently. The design principles of how an agent model works with a cellular space model are described in Hu et al. [2005]. To support the interactions between an agent and its environment (the cellular space), couplings are added between the agent and the corresponding cell where the agent locates. These couplings are dynamically added (or removed) during the simulation when the agent changes its location from one cell to another.

The *optimization* component computes firefighting resource deployment plans based on predicted fire spreading scenarios. Such plans are valuable to fire managers because they provide explicit information of what resources should be dispatched. The optimization model takes inputs, including time-indexed (e.g., every half an hour) burned areas and fire-front perimeters, from multiple runs of fire spread simulations. It also takes information of firefighting resource characteristics such as production rate, time to dispatch, and operating cost and then computes the optimal dispatch plan for containing a fire. The objective function of the SP model is to minimize the expected total cost of wildfire, which is the presuppression costs plus the expected suppression costs and fire damages. The model assumes that if the total line production of the firefighting resources exceeds the total fire perimeter, then the fire is contained. Based on this assumption, the optimization component calculates the optimal plans of what combinations of firefighting resources should be dispatched in order to contain the fire. The optimization algorithm calculates deployment plans at the operations research level without considering some realistic factors. For example, it does not specify where exactly the resources should be initially dispatched and what firefighting tactics should be used. Furthermore, the assumption of a “free burn” fire spread scenario overlooks the interaction of the fire-front with fire suppression effects.

Wildfire suppression includes many different firefighting tactics that can end up with significantly different firefighting results [NWCG 1996]. Thus there is a need for setting up fire suppression simulation that can support experimentation with various firefighting tactics. Given the dispatch plan suggested by the *optimization* component, the *firefighting agent modeling and dispatch* component is responsible for (dynamically) creating agents and dispatching them according to specific firefighting tactics for fire suppression simulation. It also supports interactive user directed dispatch. This is especially useful for fire managers to do on-the-fly “what-if” analysis by experimenting and comparing different fire suppression strategies and tactics. The component of firefighting agent modeling and dispatch utilizes two knowledge databases: a firefighting resource characteristics database including information such as the resource types and their production rates; and a firefighting strategy and tactics (e.g.,

direct attack, parallel attack, indirect attack, and their configurations) knowledge base. Such knowledge may be specified by a fire manager beforehand or interactively defined in real time as a fire manager tries out different configurations of tactics. Integrating the three components makes it possible to predict how a fire will spread, to generate firefighting resource dispatch plans based on the predicted fire, and to dynamically create and dispatch firefighting agents according to the plans and then run fire suppression simulations to evaluate them.

The last aspect of the proposed integrated simulation and optimization framework is the feedback loop from the fire suppression simulation to the optimization model. After running the fire suppression simulation, the dispatch plan proposed by the optimization model can be evaluated to be either satisfactory or unsatisfactory. The dispatch plan can be considered satisfactory if the proposed number of resources is sufficient to contain the fire within the required time. On the contrary, the dispatch plan can be considered to be unsatisfactory if the proposed number of resources cannot contain the fire, or can contain the fire but they are too many. In either case, the feedback to the optimization model would involve adding constraints to the optimization model imposing a lower bound or upper bound, respectively, on the number of resources to dispatch. This can be done iteratively until an acceptable solution is reached.

3. STOCHASTIC OPTIMIZATION MODEL FOR INITIAL ATTACK

The problem of firefighting resource management for initial attack involves making strategic decisions regarding the deployment of resources (e.g. crew, dozer, tractor plow, etc) to bases before any wildfires occur, and tactical decisions regarding which of the deployed resources at the bases to dispatch to wildfires when they occur. The goal is generally to bring the fires under control before they become too large. The difficulty of this problem comes from the uncertainty in the problem data and the combinatorial nature of which resources to deploy to the bases and then to the fires. The uncertainty stems from not knowing the exact location of where the fire will occur as well as the corresponding fire behavior (rate of spread and direction, fireline intensity, flame length, etc.) In this article, we focus on the tactical resource dispatch problem assuming that firefighting resources have already been deployed to bases for initial attack. We model this problem as a two-stage stochastic integer programming (SIP) problem with recourse, where the decision regarding which resources at given bases to dispatch to reported wildfires is made in the first stage, while recourse (corrective) actions on actual firefighting are made in the second stage based on wildfire behavior predictions.

3.1 Model Characterization

The proposed model is a stochastic extension of the deterministic model by Donovan and Rideout [2003] in which it is assumed that wildfire behavior is known, and the stochastic model by Ntamo et al. [2006], which assumes a single fire instead of multiple fires. We adopt the cost plus net value change

Table I. Example Scenario Data: Fire Perimeter and Burned Area

Hours	Perimeter (km)		Area (ha)	
	ω_1	ω_2	ω_1	ω_2
1	0.63	0.95	3.15	6.21
2	1.66	1.73	19.80	19.53
3	2.48	2.59	44.01	44.82
4	3.03	3.49	64.17	79.92
5	3.86	4.20	102.51	114.93
6	4.53	5.02	140.04	164.88

($C + NVC$) model for wildfire economics [Gorte and Gorte 1979] as in Donovan and Rideout [2003]. Thus the objective function of our model is to minimize the expected cost of containing a wildfire or multiple wildfires, that is, presuppression cost plus expected suppression cost and net value change (NVC). Presuppression cost is expenditure for acquiring (renting) firefighting resources, and suppression cost is the actual expenses for operating the resources. NVC is the net fire damage to a given area of the forest in monetary terms. Even though we use the $C + NVC$ concept, our model can still be extended to allow for other fire damage valuation criteria.

Since our model is a *fire containment* model, it deals with dispatching firefighting resources from bases to a reported wildfire to construct a perimeter around the fire. Therefore, we assume the basic principle of containment that says that if the total line production of the firefighting resources is greater than the total fire perimeter, then it can be concluded that the fire is contained [Green 1977]. It is necessary to determine whether a particular fire can be contained, since we are dealing with budgetary constraints, limited firefighting resources, and uncertainty regarding fire behavior. Given possible fire growth scenarios, a budget, and available resources, our model identifies the optimal mix of resources to dispatch with the minimum presuppression costs plus expected suppression costs and NVC . The model also determines whether the fire(s) will be contained under any given scenario.

We define a scenario as the number of wildfires on a given day with their corresponding locations and predicted fire growth characteristics: *fire perimeter* and *burned area* at specified time periods. This stochastic information is obtained from fire spread simulations. These simulations do not include direct simulation of fire containment. A sample fire growth scenario is shown in Table I. Other stochastic parameters include firefighting resource fireline production rates, arrival times to the fire and operating costs of the resources. We assume that the resources are dispatched in time period 0, which is defined as the time of dispatch after a fire has been reported. Example firefighting resource characteristics based on the Fireline Handbook [NWCG 2004] are given in Table II. The headings of the table are as follows: Arr is the arrival time of the resource to the fire location, Pre is the resource fixed (daily rental) cost, Oper is the resource operation cost, and Prod is the resource fireline construction (production) rate.

Table II. Example Firefighting Resource Characteristics

Resource	Description	Arr (hr)	Pre (\$)	Oper (\$/hr)	Prod (km/hr)
1	Dozer	2.0	300	175	0.36
2	Tractor Plow	2.5	500	150	0.45
3	Type I Crew	0.5	500	125	0.20
4	Type II Crew	1.0	600	175	0.25
5	Engine #1	1.5	400	75	0.09
6	Engine #2	1.5	900	100	0.10
7	Engine #3	1.0	600	125	0.15

3.2 Formulation

We first define the following notation and then state and give a description of the mathematical formulation.

Sets

J : Index set of firefighting resources indexed by j .

T : Index set of time periods indexed by t .

Ω : Index set of fire scenarios indexed by ω .

$\mathcal{F}(\omega)$: Index set of fires under scenario ω indexed by f .

Parameters

b_1 : pre-suppression budget.

b_2 : total budget.

h_{tf} : time period counter for fire f that takes a value of t .

p_ω : probability of occurrence of fire scenario ω .

δ_{tf}^ω : increment in fire f perimeter in period t under fire scenario ω .

v_{tf}^ω : increment in NVC for period t and fire f under scenario ω .

γ_{tf}^ω : accumulated fire perimeter up to period t under fire scenario ω .

c_j^ω : fixed rental cost of firefighting resource j under fire scenario ω .

q_j^ω : hourly cost of operating firefighting resource j under fire scenario ω .

q_s^ω : unit penalty cost for fire perimeter that is not covered under fire scenario ω .

α_j^ω : line production rate of firefighting resource j in kilometers (km) under fire scenario ω .

α_{jf}^ω : arrival time to fire f of firefighting resource j under scenario ω .

M : a large constant.

Decision Variables

x_j : $x_j = 1$ if firefighting resource j is dispatched to a fire, $x_j = 0$ otherwise.

Also, $x = [x_1, \dots, x_{|J|}]$.

y_{tf}^ω : $y_{tf}^\omega = 1$ if fire f is not contained in period t under fire scenario ω , $y_{tf}^\omega = 0$ otherwise.

w_{tf}^ω : $w_{tf}^\omega = 1$ if $y_{t-1,f}^\omega = 1$ for fire f under scenario ω , $w_{tf}^\omega = 0$ otherwise.

z_{tjf}^ω : $z_{tjf}^\omega = 1$ if containment of fire f is achieved in period t using firefighting resource j under fire scenario ω , and $z_{tjf}^\omega = 0$ otherwise.

ℓ_{tf}^ω : total line construction around fire f up to period t under scenario ω .

ℓ_{sf}^ω : total fire f perimeter that is not covered by the resources under scenario ω .

We can now state the two-stage SIP model for the wildfire containment problem (WFCP) as follows.

$$2\text{WFCP: Min } \sum_{j \in J} c_j x_j + E[\phi(x, \tilde{\omega})] \quad (1a)$$

$$\text{s.t. } \sum_{j \in J} c_j x_j \leq b_1 \quad (1b)$$

$$x_j \in \{0, 1\}, \forall j \in J \quad (1c)$$

where for each scenario (outcome) $\omega \in \Omega$ of $\tilde{\omega}$,

$$\phi(x, \omega) = \text{Min } \sum_{f \in \mathcal{F}(\omega)} \left\{ \sum_{t \in T} \sum_{j \in J} h_{tf} q_j^\omega z_{tjf}^\omega + \sum_{t \in T} v_{tf}^\omega w_{tf}^\omega + q_s^\omega \ell_{sf}^\omega \right\} \quad (2a)$$

$$\text{s.t. } \sum_{f \in \mathcal{F}(\omega)} \sum_{j \in J} \sum_{t \in T} q_j^\omega h_{tf} z_{tjf}^\omega \leq b_2 - \sum_{j \in J} c_j x_j \quad (2b)$$

$$\sum_{f \in \mathcal{F}(\omega)} \sum_{t \in T} z_{tjf}^\omega \leq x_j, \forall j \in J \quad (2c)$$

$$\sum_{j \in J} (h_{tf} - \alpha_{jf}^\omega) \alpha_j^\omega z_{tjf}^\omega - \ell_{tf}^\omega = 0, \forall t \in T, f \in \mathcal{F}(\omega) \quad (2d)$$

$$\sum_{t \in T} \ell_{tf}^\omega - \sum_{t \in T} \delta_{tf}^\omega w_{tf}^\omega + \ell_{sf}^\omega \geq 0, \forall f \in \mathcal{F}(\omega) \quad (2e)$$

$$\gamma_{tf}^\omega w_{tf}^\omega - \ell_{tf}^\omega - M y_{tf}^\omega \leq 0, \forall t \in T, f \in \mathcal{F}(\omega) \quad (2f)$$

$$w_{tf}^\omega - y_{t-1,f}^\omega = 0, \forall t \in T, f \in \mathcal{F}(\omega) \quad (2g)$$

$$z_{tjf}^\omega \in \{0, 1\}, y_{tf}^\omega \in \{0, 1\}, y_{0f}^\omega = 1, w_{tf}^\omega \in \{0, 1\}, \ell_{sf}^\omega \geq 0, \ell_{tf}^\omega \geq 0, \forall t \in T, j \in J, f \in \mathcal{F}(\omega). \quad (2h)$$

In 2WFCP, the first stage objective function (1a) is to minimize the presuppression costs plus the expected suppression cost and *NVC* of the burned area. A standard assumption in stochastic programming is that the probability distribution of $\tilde{\omega}$ is assumed to be known or given. Constraint (1b) is a presuppression budgetary constraint on firefighting resources fixed/rental costs. Constraints (1c) are binary restrictions on the first-stage decisions. Given the mix of firefighting resources determined in the first stage, for a given daily fire scenario $\omega \in \Omega$, the second-stage objective function (2a) minimizes the sum of the suppression cost, *NVC*, and a penalty cost for the uncovered perimeter if a fire is not contained. If the dispatched resources cannot contain a fire, the variable ℓ_{sf}^ω takes a positive value. In our computational results we calculated the penalty q_s^ω as 100 times the largest coefficient in the objective function. This value was sufficient to drive ℓ_{sf}^ω to zero if the fire cannot be contained in that scenario.

The budgetary constraint on the total presuppression and suppression costs regarding the dispatched resources is given by (2b). Constraint (2c) ensures that the resource used in any time period is actually dispatched in the first stage and is used to fight one fire. The total fireline produced by the dispatched resources in time period t from time period 0, ℓ_{tf}^ω , is computed in constraint (2d) for a given fire. Constraint (2e) indicates that, for a given $\omega \in \Omega$, ℓ_{tf}^ω must exceed

the total fire perimeter at some time period $t \in \mathcal{T}$. Otherwise, the fire is not contained and the variable ℓ_{sf}^ω is forced to take a positive value.

Constraint (2f) is a logic constraint on whether the fire is contained in the time period $t \in \mathcal{T}$. If ℓ_{tf}^ω is less than the total fire perimeter in time period t then the fire is not yet contained. Consequently, the indicator decision variable y_{tf}^ω takes on a value of 1, otherwise it takes on a value of either 0 or 1. If the fire is contained, the second stage objective function forces y_{tf}^ω to take on a value of 0 to minimize the total cost. We note that the constant M can be defined as $M \geq \text{Max}_{\omega \in \Omega, t \in \mathcal{T}} \{\gamma_{tf}^\omega - \ell_{tf}^\omega\}$. In constraint (2g), w_t^ω is defined as a one-time lagged variable on y_{tf}^ω to ensure that increments of fire perimeter growth and damage are included for time periods during which fire containment is not achieved, and for the time period when it is achieved. Also note that constraint (2f) links two different time periods. Constraint (2h) imposes binary and nonnegativity restrictions on the second-stage decision variables. The initial condition $y_{0f}^\omega = 1$ starts the model with the fire having been ignited by time period $t = 0$.

We should point out that regardless of the fire spread simulation scenario data that is given to the optimization model, model WFCP remains feasible due to the slack variables ℓ_{sf}^ω , which appear in constraints (2e) and penalized in the objective function (2a). Since these slack variables give the total fire perimeter that is not covered by the resources under scenario ω , any dispatch solution with $\ell_{sf}^\omega > 0$ implies that based on the optimization model, the fire cannot be contained under that scenario. However, such a solution can actually be “feasible” to the fire suppression simulation, that is, the fire can actually be contained. So our optimization model allows for generating such solutions for further evaluation by the fire suppression simulation. We also note that problem (1-2) is a multiperiod SIP problem with *random recourse*, that is, the constraint matrix in the second-stage subproblem depends on ω . Since we are dealing with finite support for the random variable $\tilde{\omega}$, we can rewrite problem (1-2) in extensive form (deterministic equivalent form) as a large-scale mixed-binary program that can be directly given to a mixed-integer programming solver. However, if $|\Omega|$ is very large, decomposition methods for SIP with random recourse [Ntamo 2008] can be used to solve 2WFCP.

4. FIRE SUPPRESSION SIMULATION MODEL

Fire suppression is a process for firefighting agents to construct a fireline to suppress (or contain) a burning fire. Three types of fire suppression strategies, *direct attack*, *parallel (indirect) attack*, and *indirect attack*, are considered. Direct attack refers to the strategy in which fireline is constructed on the flaming fire-front, the region where combustible fuels are igniting. Two specific direct attack tactics are direct head attack and direct tail attack where the attacks start from the head and tail of the fire respectively. Parallel (indirect) attack refers to the strategy in which fireline is constructed parallel to, but at a safe distance (offset) away from, the fire perimeter. This is usually applied when the fire is intensive and fast spreading, thus having the potential for causing serious injuries or fatalities to the firefighters. Indirect attack refers to the strategy in which fireline is constructed according to a predetermined route. Besides

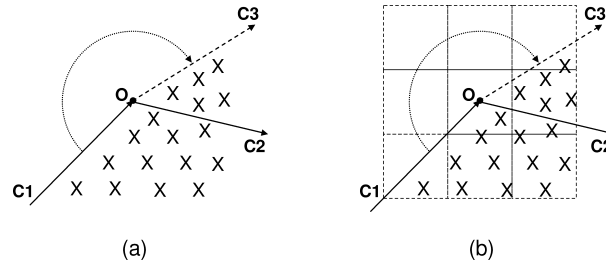


Fig. 2. The “predict-and-scan” schema to choose the next suppression cell.

these three strategies, firefighting resources may also be divided into different groups working concurrently on different fireline segments. More discussions about these different tactics can be found in the fire suppression handbook [NWCG 1996].

In developing the fire suppression simulation, we make the following two assumptions that are common for all three attack strategies. These assumptions are made because of the discrete nature (discrete space and discrete event) of the fire spread simulation model. (1) In a cellular space model where the space is divided into discrete cells, a firefighting agent can only proceed, that is, construct the fireline, from center to center between two neighbor cells. (2) The effect of fire suppression is a Boolean effect (true or false). When an agent reaches the center of a cell, the cell is considered suppressed. Otherwise, the cell is treated as unsuppressed and can ignite its neighbors.

Simulating fire suppression mainly deals with how to simulate the dynamical process of fireline construction that is carried out by firefighting agents. Each agent, after completing a fireline segment, needs to decide where to proceed for constructing the next fireline segment. This decision is either based on a pre-defined route (such as in indirect attack) or on the dynamical behavior of fire spread (such as in direct attack and parallel attack). In this work, since an agent can only move from a cell to a neighboring cell in a discrete fashion, the modeling of firefighting agent concerns how an agent chooses an appropriate neighboring cell as the destination cell to construct the next fireline segment. Below we present the agent models for direct attack, parallel attack, and indirect attack respectively.

4.1 Direct Attack

In direct attack, agents build a fireline along the fire-front where cells are burning. Since it takes time for a fireline segment to be constructed, a burning cell may ignite its neighboring cells even if it is being attacked by an agent. Thus in choosing which neighboring cell to construct the next fireline segment, an agent needs to ensure that the to-be-constructed fireline segment will be completed before any neighboring cells “outside” this fireline segment are ignited. In the agent model, this is achieved by a “predict-and-scan” schema that is illustrated in Figure 2 and described next.

The basic idea of the “predict-and-scan” schema is that an agent needs to predict how far the current fire can spread-based on how fast the agent itself can

finish the next fireline segment, and then uses the predicted fire-front as a guidance to decide the direction for constructing the next fireline segment. This is illustrated in Figure 2(a), where O represents the agent's current location, line segment C1-O represents the already completed fireline, line segment O-C2 represents the current fire-front, and line segment O-C3 represents the predicted fire-front. The “look-ahead” time window that is used for fire spreading prediction is based on how fast the agent can finish the next fireline segment. This is calculated by dividing the segment distance by the agent's production rate. After predicting the fire spread, the agent then chooses the predicted fire-front (e.g., O-C3 in Figure 2(a)) as the direction for constructing its next fireline segment. Because this is the predicted fire-front, it ensures that during the time when the agent builds the chosen fireline segment, no area “outside” the fireline perimeter (e.g., the top left area in Figure 2(a)) will be ignited. To detect the predicted fire-front, the agent applies a “scan” process. Specifically, starting from the completed fireline segment and taking the circular direction that goes away from the burning side of that segment, the agent scans around until it meets the first burning area. This is illustrated by the dashed circular arrow in Figure 2(a). Figure 2(b) illustrates the “predict-and-scan” schema in a cellular space. In this figure, the completed fireline segment comes from the southwest neighbor of the current cell (both the southwest cell and the current cell are thus suppressed). The east neighboring cell represents the current fire-front and the northeast neighboring cell represents the predicted fire-front. By “scanning” its neighboring cells, the agent chooses the northeast neighboring cell as the cell for constructing the next fireline segment.

Based on the design idea we have described, a two stage “predict-and-scan” schema is developed for the agents in direct attack. The two stages are needed because the distance for a diagonal fireline segment and that for a non-diagonal fireline segment are different, that is, the distance to a diagonal cell is $\sqrt{2}$ as long as that to a nondiagonal neighboring cell. This difference results in different time that is needed for an agent to construct the fireline segments. Because of this, two different “look-ahead” time windows, and thus two stages of prediction, should be used to predict fire spread in order for the agent to decide the next fireline segment. The algorithm that implements the two stage “predict-and-scan” schema is described below. Specifically, after an agent reaches the center of a cell, the agent goes through the following steps to choose a neighboring cell for constructing the next fireline segment. Let T_{la} denote the look-ahead time window (seconds), d denote the size of a cell (meters), and α be the agent's production speed (meters per second or m/s) for building a fireline.

Begin choose-neighbor-cell in direct attack

Step 1. Calculate the time for building a fireline to a nondiagonal neighboring cell: $T_{la} = \frac{d}{\alpha}$.

Step 2. Use T_{la} as the look-ahead time window to predict the fire spreading situation of the neighboring cells. Mark them as unburn, burning, burned, or suppressed.

Step 3. Apply the “scan” process described above to scan its neighboring cells until meet the first burning cell. Two situations may occur: (i) If this cell is a non-diagonal cell, the cell is chosen as the destination cell. Go to Step 7. (ii) If this cell is a diagonal cell, aborts the scan and starts the second “prediction-and-scan” stage (Step 4).

Step 4. If a destination cell is chosen, go to Step 7. Otherwise, calculate the time for building a fireline to a diagonal neighboring cell: $T_{la} = \sqrt{2}(\frac{d}{\alpha})$.

Step 5. Use T_{la} as the look-ahead time window to predict the fire spreading situation of the neighboring cells. Mark them as burning, burned, unburn, or suppressed.

Step 6. Apply the “scan” process described above to scan its neighboring cells until meet the first burning cell. Choose this cell (diagonal or non-diagonal) as the destination cell.

Step 7. Hold for a period of time $\frac{d}{\alpha}$ for a nondiagonal destination cell and $\sqrt{2}(\frac{d}{\alpha})$ for a diagonal destination cell. After that time elapses (meaning the fireline is constructed), change the agent’s location and also send a “suppressed” message to the destination cell. Reiterate this process starting from Step 1.

end choose-neighbor-cell in direct attack

The algorithm that is used to predict fire spread is the fire spreading simulation itself. Specifically, an agent creates a new cell space model that duplicates the local area of the “original” cell space. The states of the cells in this new cell space are initialized to the current states of the corresponding cells in the original cell space. This new cell space model is then simulated until the given look-ahead time window is reached. Creating a cell space model that represents only the local subarea, instead of the entire cell space, for simulation is due to performance considerations. By duplicating the states of all local cells, the created cell space incorporates all the information of the local area. This is generic for various situations such as different fire shapes and/or multiple fires. Also note that in the two stage “predict-and-scan” schema, for simplicity we use the same “look-ahead” predicting time window for all nondiagonal (or diagonal) cells. However in an area with non-uniformed fuel models (or terrain data), the fireline construction time to different cells can be different and is dependent on the fuel model (or terrain data) of the cell (see, e.g., a discussion in Caballero [2002]). To have more precise predictions in such cases, the two-stage “predict-and-scan” schema can be extended to a multistage “predict-and-scan” schema to account for non-uniform fuel models (or terrain data).

4.2 Parallel Attack

In parallel attack, agents build a fireline parallel to the fire-front perimeter and maintain a fixed safe distance to it. One can view direct attack is a special case of parallel attack with safe distance being 0. Based on this view, the same algorithm, that is, the “predict-and-scan” schema, used in direct attack is

employed in parallel attack to compute the fireline path. The major difference is that an agent in direct attack finds the first cell that is burning as the destination cell when it “scans” its neighboring cells. But in parallel attack, the agent chooses a neighboring cell whose distance to the closest fire-front equals to (or is close to) the predefined safe distance as the destination cell. Specifically, after an agent reaches the center of a cell, the agent goes through the following steps to choose a neighboring cell for building the next fireline segment:

Begin choose-neighbor-cell in parallel attack

Step 1. Calculate the time for producing a fireline to a diagonal neighboring cell, $T_{la} = \sqrt{2}(\frac{d}{\alpha})$.

Step 2. Use T_{la} as the look-ahead time window to predict the fire spreading situation of the agent’s distance bounded neighboring cells. Here the distance bounded neighboring cells are those cells whose cell-distances (distance based on difference of cell IDs) are less than or equal to $\lceil \frac{D_{safe}}{d} \rceil + 1$, where D_{safe} is the desired safe distance. Based on the prediction, mark the cells as burning, burned, unburn, or suppressed.

Step 3. Apply the same “scan” process as described in direct attack to scan the agent’s direct neighboring cells. Let D_{fire} denote the distance from this cell to the closest fire-front. For each cell, calculate D_{fire} and choose the first cell (diagonal or non-diagonal) whose $D_{fire} = D_{safe}$ (or $D_{fire} \approx D_{safe}$) as the destination cell to construct the next fireline segment.

Step 4. Hold a period of time equal to $\frac{d}{\alpha}$ for a nondiagonal destination cell and $\sqrt{2}(\frac{d}{\alpha})$ for a diagonal destination cell). After that time elapses (meaning the fireline is constructed), change the agent’s location and also send a “suppressed” message to the destination cell. Reiterate this process starting from Step 1.

end choose-neighbor-cell in parallel attack

Two things are worth mentioning here. First, the firefighting agents in parallel attack use a single stage instead of two-stage, “predict-and-scan” schema to calculate the fireline path. This is a simplified treatment and is used in the current implementation. Because the longer distance (i.e., the distance to a diagonal cell) is used to calculate the look-ahead time window for prediction, this treatment guarantees no burning cell will be left out of the fireline perimeter. Second, an agent in parallel attack needs to predict the fire spreading situation of its distance bounded neighboring cells. The value of the “distance bound” is dependent on the desired safe distance in parallel attack. This is different from that in direct attack, which only predicts fire spreading of the agent’s direct neighboring cells.

4.3 Indirect Attack

In indirect attack, agents build a fireline according to a predetermined route. Such a predetermined route can be explicitly specified by a fire manager, or

be generated from some algorithm that considers factors such as terrain, fire spreading behavior, and available firefighting resources. How to generate an effective route for indirect attack is out of the scope of this article. Given such a predetermined route, below we describe how a firefighting agent proceeds in indirect attack. Because of the existence of a predetermined route, the design of firefighting agents in indirect attack is simpler than those in direct attack and in parallel attack. Specifically, each agent has a copy of the predetermined route. After an agent reaches the center of a cell, the agent uses its current location to check the route, and then chooses a neighboring cell that is consistent to what the route suggests. It is assumed that an agent will always follow the predetermined route in an indirect attack, independent of how the fire actually spreads.

4.4 Multiple Agents

It is common for multiple firefighting resources to work together to suppress a fire. In our design, dependent on the group configurations of the multiple resources, we handle the multiple resources in two different ways. First, if multiple resources work as a single group on the same fireline segment, they are treated as a single agent with its fireline production rate being the aggregated production rates of all the resources. For example in a direct tail attack, if two resources stay together and build a fireline in the same direction, the effect of fire suppression by these two resources is simulated using a single agent whose fireline production rate equals to the sum of those of the two resources. On the other hand, if multiple resources work as different groups on different fireline segments (either on different locations or following different directions), they are treated as independent agents without influencing each other. For example, if one resource works in the clockwise direction and the other in the counter-clockwise direction, or if one works in the head of the fire and the other in the tail, then the two resources work independently. In general, multiple resources are divided into several groups, each of which works on its own fireline segment. In this case, the agents that work on the same fireline segment are simulated as a single agent, and different groups (simulated by different agents) work concurrently. In a successful fire containment, the different fireline segments constructed by different agents will eventually merge to close the fire area. In the current implementation, an agent stops proceeding whenever it meets a fireline constructed by other agents (this means the two firelines merge).

4.5 Dispatch of Firefighting Agents

Different firefighting resources may be dispatched to different locations of the fire-front. They can work as different groups, and use different firefighting tactics. Furthermore, these resources typically have different dispatch time due to factors such as mobility and firebase locations where the resources come from. In fire suppression simulation, a dispatch model is responsible for sending the firefighting agents to a given wildfire. Specifically, as the dispatch time of a resource arrives, the dispatch model dynamically creates an agent based on its dispatch configuration. This configuration includes information of group ID,

dispatch time, dispatch location, firefighting direction (clockwise or counter-clockwise), tactic types (direct attack, indirect attack, parallel attack), and production rate. The group ID indicates if this resource joins an already dispatched group (if group ID matches an existing group ID) or starts a new group. In the former, the existing group's production rate is increased by this resource's production rate. In the later, the agent creates a new group and starts to work from the location specified. The dispatch configuration is an essential part of fire suppression simulation and is specified by a fire manager before running the simulation.

5. EXPERIMENTAL RESULTS

This section presents the results from three experiments. The first experiment demonstrates different firefighting tactics by running simulations with different firefighting configurations. The second experiment illustrates how the fire spread simulation, firefighting resource optimization, and fire suppression simulation can work together to provide valuable information for wildfire containment. The third experiment demonstrates the framework in a more realistic setting using real GIS data. In the last two experiments, all fire suppression simulations consider only direct attack where firefighting resources construct the fireline directly on the fire-front. It is worthwhile to point out that in a real fire, direct attack may not be applied due to the high fireline intensity or flame length. In such cases, parallel and indirect attack are the only tactics that can be employed. Some rules about how to select attack tactics can be found in Roethermel and Rinehard [1983], Andrews [1986], and Caballero [2002]. This article has not taken into account such rules and assumes direct attack can be generally applied. In all the following simulations, the maximum simulation time is set to 6 hours, which is the assumed time period for initial attack in this article.

5.1 Simulation Results with Different Firefighting Tactics

This experiment varies the configurations of firefighting resources and shows how they lead to different fire suppression results. Specifically, it compares firefighting tactics from three aspects: (1) different types of attack including direct attack, parallel attack, and indirect attack; (2) different direct attack methods including direct head attack, direct tail attack, and combined head and tail attack; and (3) different group configurations of firefighting resources. All simulations use a 200×200 cellular space. For display purpose, only the central part of the space is shown in the figures (the center cell has coordinates (100, 100)). Each cell has size 30×30 meters, 0 slope and 0 aspect, and fuel model 10, which represents timber (litter and understory). A constant wind is applied with wind speed 6 miles/hour and direction from south to north. Fire is first ignited at cell (100, 77) at time 0. In all the figures shown in the following, the green color means a cell is unburn, red means burning (including the "free burn" fire perimeter described below), black means burned, yellow means suppressed while burning, and blue means suppressed while unburn. Besides showing the suppressed fire, all figures also display the "free burn" fire

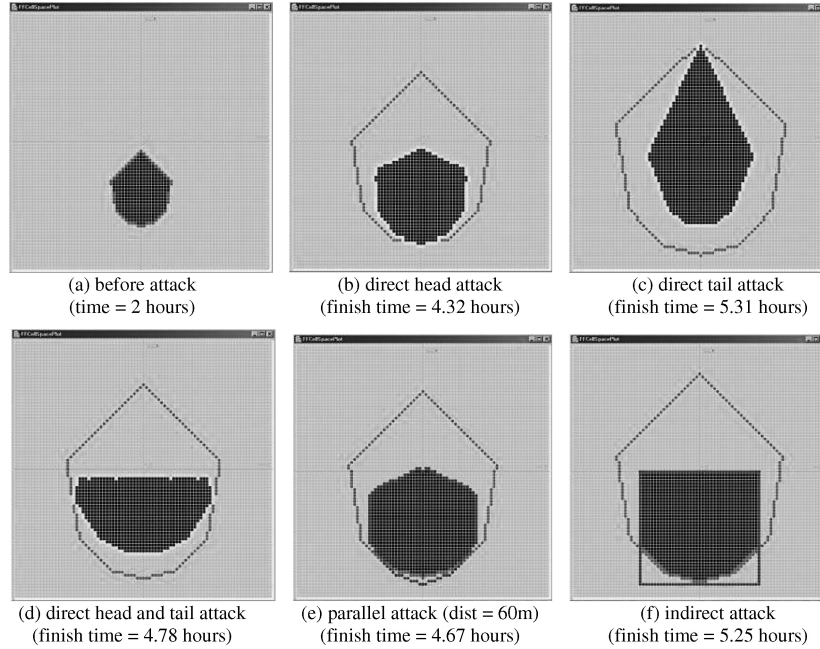


Fig. 3. Fire suppression with different tactics.

perimeter (i.e., if no fire suppression is applied) up to the same time when the fire is suppressed or the maximum simulation time is reached. The suppressed fire shape is displayed by the inner perimeter in yellow or blue color, while the “free burn” fire shape is displayed by the outer perimeter in red color.

The first set of simulations (shown in Figure 3) compares different direct attack methods as well as parallel attack and indirect attack. Let T_s denote the starting time of fire suppression, P be the overall production rate of all agents, N be the total number of agents, $A_i (i = 1, \dots, N)$ be the i -th agent, and p_i be the production rate of agent A_i . In all the simulations shown in Figure 3, fire suppression starts at $T_s = 2$ hours, and agents have overall production rate $P = 0.4$ m/s (1.44 km/hour). Figure 3(a) shows the fire shape at time T_s when the fire suppressions start. Figure 3(b) and 3(c) show the results of direct head attack and direct tail attack, where two agents are used (one moves clockwise and the other moves counterclockwise) and each agent has production rate $p_i = 0.2$ m/s. In direct head attack, the two agents start from the same cell (100,98), while in direct tail attack they start from cell (100,72). Figure 3(d) shows the result of a combined head and tail attack. This attack combines the configurations of direct head attack and direct tail attack mentioned above and employs four agents, each of which has production rate $p_i = 0.1$ m/s. Figure 3(e) shows the result of a parallel attack and Figure 3(f) shows an indirect attack. These two attacks use two agents with $p_i = 0.2$ m/s. In the parallel attack, the two agents start from cell (100,100) and maintain a parallel distance of 2 cell size (each cell size = 30m). In the indirect attack, the two agents start from cell (100,100) and follow a predefined route in a rectangle shape as shown in

Table III. Results of Direct Attacks, Parallel Attack, and Indirect Attack

Firefighting Tactic	Suppression Time (hrs)	Finish Time (hrs)	Fire Perim. (km)	Burned Area (ha)
Before attack		2.0	2.2	36.09
Direct head attack (2 agents, prod. rate 0.20 each)	2.32	4.32	3.23	71.64
Direct tail attack (2 agents, prod. rate 0.20 each)	3.31	5.31	4.77	123.30
Direct head & tail attack (4 agents, prod. rate 0.10 each)	2.78	4.78	3.89	89.37
Parallel attack (2 agents, prod. rate 0.20 each)	2.67	4.67	3.79	93.6
Indirect attack (2 agents, prod. rate 0.20 each)	3.25	5.25	4.68	129.87

Figure 3(f). Table III shows the fire suppression time (the time duration for all agents to finish the suppression), finish time (suppress time plus the starting time T_s), fire perimeter (the perimeter length of the suppressed fire), and burned area (the area that is surrounded by the perimeter) for the five attacks described above. The total fireline length constructed by the agents can be approximated by multiplying the suppression time with the overall production rate (0.4 m/s).

Figure 3 and Table III show that all the five attacks succeeded in suppressing the fire within 6 hours. However, their fire shapes and perimeters/burned areas are different. The finish time also varies from 4.32 hours for direct head attack to 5.31 hours for direct tail attack. Considering the three methods of direct attack, one can see that direct head attack is most effective and direct tail attack is least effective. This is because the head of a fire spreads much faster than the tail does. Thus allocating the firefighting resources firstly at the head of a fire as in direct head attack can result in less suppress time and smaller fire perimeter and burned area. However, it should be pointed out that direct head attack may be more effective most of the time but it may also be more risky if behavior changes, for example, wind speed increases or direction shifts such that firefighting forces are threatened with becoming overwhelmed. Moreover, head attack can be unsafe for very severe wildfires. The data in Table III shows that with the same $P = 0.4$ m/s, the direct head attack finishes 1 hour earlier than the direct tail attack, and results in significant less fire perimeter (32% less) and burned area (42% less). For the combined head and tail attack, Figure 3(d) shows that the head portion of the fireline roughly ends up with a straight line. This is because each agent has only 0.1 m/s production rate, which barely allows them to compete with the fire spreading speed. This observation of the race between fire suppression and fire spreading brings out an interesting fact: a head attack is effective only when the resource's production rate is fast enough to quickly contain the fire. In situations where fire spreads fast (such as due to high wind speed) and firefighting resources are limited, head attack is not desirable because the fast spreading fire can bypass the effort of fire suppression and potentially surround the firefighting resources. Like direct attack, both parallel attack and indirect can start from the head, tail, combined

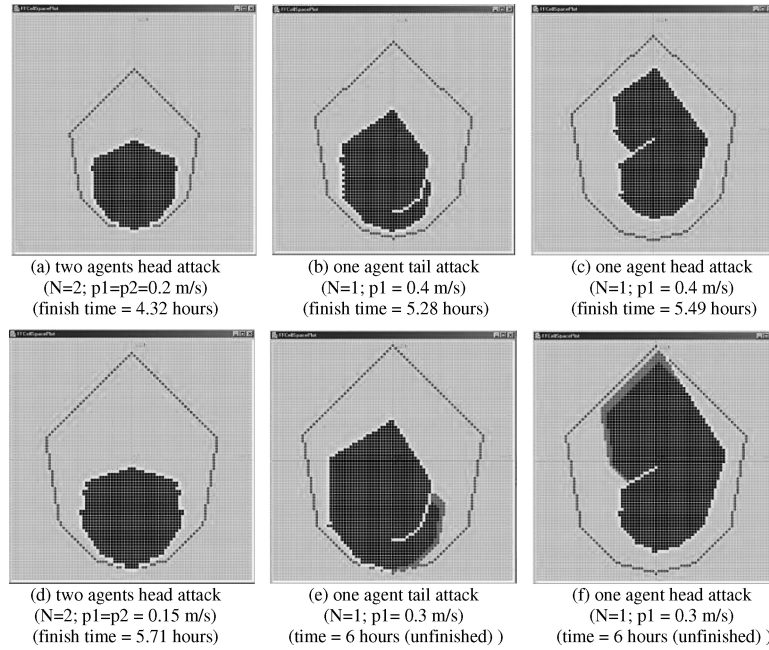


Fig. 4. Fire suppression with different group configurations.

head and tail, or other areas of the fire. Figure 3(e) and 3(f) shows a parallel attack and an indirect attack starting from the head of the fire. It can be seen that the parallel attack results in a similar, but larger, fire shape as the direct head attack, while the indirect attack follows a pre-defined route and is independent of how the fire spreads. In general, compared to their corresponding direct attack, parallel attack and indirect attack result in longer suppression time and larger fire perimeters/burned areas. Table III shows that the parallel head attack with two cell size parallel distance results in about 15% more fireline than the direct head attack.

The second set of simulations (shown in Figure 4) compares the fire suppression results of different group configurations of firefighting resources. Same as before, the starting time of all the fire suppressions $T_s = 2$ hours. Figures 4(a), 4(b), and 4(c) show three simulations where the overall production rate $P = 0.4$ m/s. In Figures 4(a), the firefighting resources are equally divided into two groups (each group has production rate 0.2 m/s) for direct head attack. This is the same simulation as shown in Figure 3(b). In Figures 4(b) and 4(c), all resources work as one group (thus one agent) with production rate 0.4 m/s. Figure 4(b) corresponds to a tail attack starting from cell (100,72), and Figure 4(c) corresponds to a head attack starting from cell (100,98). Figures 4(d), 4(e), and 4(f) show three simulations where the overall production rate P is reduced to 0.3 m/s. Except for this change, their other configurations are the same as those in Figures 4(a), 4(b), and 4(c) respectively. Table IV shows the fire suppression time, finish time, fire perimeter, and burned area for the six simulations described above.

Table IV. Results for Different Group Configurations

Number of Agents	Suppression Time (hrs)	Finish Time (hrs)	Fire Perim. (km)	Burned Area (ha)
Direct head attack (2 agents, prod. rate 0.20 each)	2.32	4.32	3.23	71.64
Direct head attack (1 agent, prod. rate 0.40 each)	3.49	5.49	4.64	122.31
Direct tail attack (1 agent, prod. rate 0.40 each)	3.28	5.28	3.89	99.99
Direct head attack (2 agents, prod. rate 0.15 each)	3.71	5.71	3.81	93.78
Direct head attack (1 agent, prod. rate 0.30 each)	4.0 [†]	6.0 [†]	5.81	194.13
Direct tail attack (1 agent, prod. rate 0.30 each)	4.0 [†]	6.0 [†]	4.71	148.41

[†]Fire not contained.

From Figure 4 and Table IV, one can see that when the total production rate equals 0.4 m/s, all the three group configurations (Figures 4(a), 4(b), and 4(c)) succeeded in containing the fire within 6 hours. However, both the one-agent head attack and tail attack result in longer suppress time (About 1 hour longer) and larger fire perimeters and burned areas than the two-agent head attack. This is because when all firefighting resources work as one group on one side of the fire, the fire expands unbounded on the other side and thus requires more time to contain. The one-agent tail attack is more effective than the head attack because the uncontained fire in the tail attack spreads much slower than that in the head attack. Note that in both Figures 4(b) and 4(c), the yellow fireline inside the burned area is the constructed fireline that is surrounded by the uncontained fire. Figures 4(b) and 4(c) also show that even though fire is expanded from the tail or head, the one agent was still able to contain the fire due to its high production rate. However, this did not happen as the production rate was reduced to 0.3 m/s. Figures 4(d), 4(e), and 4(f) show that when $P = 0.3$ m/s, the two-agent head attack was able to contain the fire, but needs about 1.5 hours longer than when $P = 0.4$ m/s. However, none of the one-agent head attack and one-agent tail attack succeeded in containing the fire within 6 hours. Figures 4(e) and 4(f) clearly show how the fires grew from the firelines constructed by the agents.

These simulations, although carried out using simplified settings such as unchanged weather condition and uniform fuel model, illustrate two important features of wildfire suppression. First, for a given set of firefighting resources, fire suppression can take different forms according to different firefighting tactics. Second, different firefighting tactics can potentially give significantly different fire suppression results. Based on the experimental results, some empirical conclusions can be made. For example, when direct attack is considered, it is more effective to start fire suppression from the head area of the fire that spreads the fastest (if it is safe to apply head attack). Furthermore, in an effort of head attack, two agents are more effective than one agent because two agents can cover the head area in a more effective manner even with the same total

production rate. These observations motivate us to investigate how to evaluate the resource dispatch plans generated by the optimization algorithm when using specific firefighting tactics.

5.2 Results from Integrated Simulation and Optimization

This experiment demonstrates how fire spread simulation, firefighting resource optimization, and fire suppression simulation can work together and provide valuable information for suppressing a wildfire. The experiment starts from fire spread simulations that generate predicted fire spreading scenarios. Given these scenarios, the optimization component then computes the resource dispatch plans that can contain the fires based on the resources that are available. Finally, fire suppression simulations evaluate the resource dispatch plans by running simulations with different firefighting tactics and stochastic variables.

This experiment considers a uniform 200×200 cellular space area with cell size 30×30 meters, fuel model of 10, slope of 0 and aspect of 0. It involves two fire spread simulations for 6 hours that represent two fire spreading scenarios ω_1 and ω_2 . In each scenario, wind speed and wind direction are randomly generated every half an hour for six hours. The wind speed is randomly generated from *uniform*(5, 25) miles/hour while the wind direction is sampled from *uniform*(30, 120). We use the conventional sense for the wind direction, that is, the direction that the wind is coming from. The fire is ignited at the center (cell (100, 100)) of the study area at time $t = 0$. During the simulation run, we record the time-indexed fire perimeters and burned areas at the end of every one hour for a six hour prediction of fire spread. These results are the ones reported in Table I. Figure 5(a) shows the final fire shape at $t = 6$ hours for scenario ω_2 .

Next, the time indexed fire perimeters and burned areas (given in Table I) are fed into the optimization model 2WFCP. The firefighting resource characteristics given in Table II were used for this experiment. Based on these data, this instance of 2WFCP was solved using the CPLEX MIP solver [ILOG 2003] to get an optimal dispatch plan. The optimal solution is to deploy the following firefighting resources: Dozer, TPlow, CrewI, CrewII, Eng1, and Eng3. In both scenarios, the fire is contained in time period $t = 4$ hours. For scenario ω_1 the resources dispatched to the fire are Dozer, TPlow, CrewI, CrewII, and Eng1, while for scenario ω_2 all six resources are dispatched. The fireline that can be constructed at the end of four hours for scenario ω_1 is 3.07 km, which is greater than the fire perimeter of 3.03 km. For scenario ω_2 , the constructed fireline is 3.52 km, which is greater than the fire perimeter of 3.49 km (see Table I).

Finally, given the dispatch plans we have described, we ran fire suppression simulations to evaluate the plan. In the following we use the dispatch plan for scenario ω_2 to demonstrate firefighting simulation since this scenario dispatched all the resources. As mentioned before, many different firefighting tactics can be set up and tested. In the following, we only consider the case where all resources working as one group (simulated by one agent) for direct attack in a clockwise direction. Since total 6 resources are dispatched and their arriving time is different, the agent's production rate dynamically increases

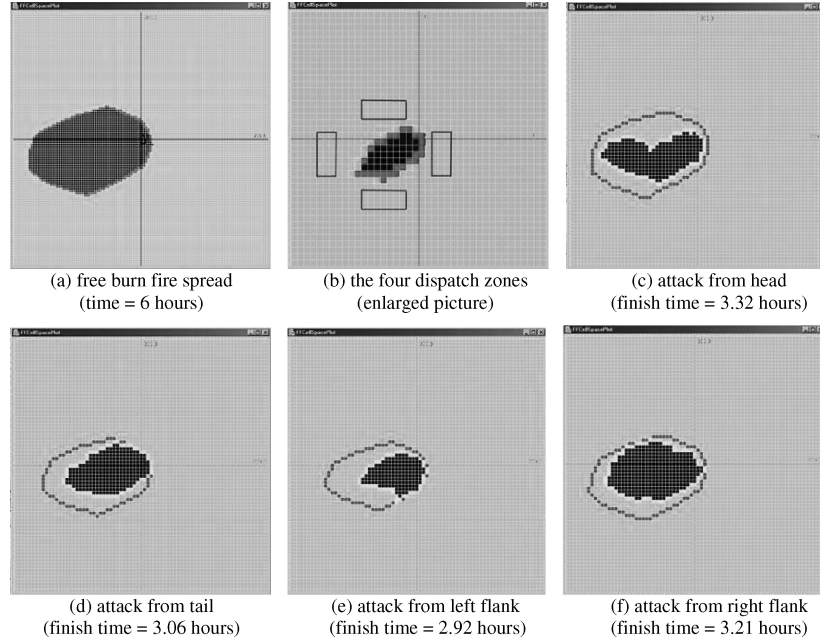


Fig. 5. Fire suppression from different dispatch locations.

when a new resource arrives (join the group). For example, based on the dispatch plan and resources' characteristics, the firefighting agent starts the fire suppression at $t = 0.5$ hours with production rate 0.2 km/hr (since resource CrewI arrives the earliest at $t = 0.5$ hours). The agent maintains that rate until resource CrewII arrives at $t = 1$ hour, when the agent's production rate increases to 0.45 km/hr. This process continues until all resources arrive and the agent's production rate reaches its maximum level. To give a comprehensive evaluation of the firefighting dispatch plan, we ran three sets of simulations that investigate three aspects of the dispatch plan: *initial dispatch location*, *arriving time*, and *production rate*.

The first set of simulations investigates how the initial dispatch location can affect the fire suppression result for the given dispatch plan. In these simulations, we defined four dispatch zones where the agent was initially dispatched and started the fire suppression. Figure 5(b) shows these four zones (displayed in blue rectangles) in relation to the fire shape at $t = 2$ hours. These zones roughly correspond to the head (left zone), tail (right zone), left flank (bottom zone), and right flank (top zone), of the fire. We note that the four zones are defined based on the 2 hour fire shape instead of the 0.5 hour fire shape because the agent's initial production rate is very low. With this low rate, dispatching the agent too close to the fire front will cause the agent to be surrounded by the fast spreading fire. Also note that due to the irregular fire shape, simulation results from these four zones may be slightly biased. For example, the left and right flank zones are not exactly symmetric and the fire reaches the left flank zone quicker because of its spreading direction. Each zone has 21 cells. For each

Table V. Results of Fire Suppression for Different Dispatch Areas

Dispatch Area	Suppression Time (hrs)	Finish Time (hrs)	Fire Perim. (km)	stdev	Burned area(ha))	stdev
Head area	2.82	3.32	2.39	0.130	28.51	4.23
Tail area	2.56	3.06	2.05	0.126	24.61	1.35
Left flank area	2.42	2.92	1.77	0.116	17.95	1.84
Right flank	2.71	3.21	2.24	0.081	32.04	2.03

zone, we ran 21 simulations, each of which started the fire suppression from a different cell in the zone.

Table V shows the average fire suppression time, finish time, fire perimeter, burned area, and standard deviation of the fire perimeter and burned area based on the 21 simulations for each zone. Figures 5(c), 5(d), 5(e), and 5(f) show a typical suppressed fire shape (the inner yellow perimeter) as well as its “free burn” fire shape (the outer red perimeter) for attacks from the head, tail, left flank, and right flank zones respectively. Note that for display purpose, these figures are not in the same resolutions as Figures 5(a) and 5(b). From Figure 5 and Table V, one can see that the initial dispatch location where fire suppression starts is an important factor to consider in suppression a fire. Different initial dispatch zones give different fire suppression results. Among them, the left flank zone is most effective because the agent covered the head of the fire in an early stage and prohibited the fire from escape. The head zone is least effective because part of the fire in the head area is expanded. Specifically, the head zone results in 16% more suppress time, 35% more fire perimeter, and 58% more burned area than the left flank zone. Table V also shows that the standard deviation of the fire perimeter for a particular zone is small. This means changing the specific dispatch location within a zone does not bring significant change of the fire suppression result. Thus it makes sense to roughly specify different zones, such as the ones corresponding to the head, tail, left flank, and right flank of a fire, for dispatching firefighting resources to suppress a fire.

The second set of simulations investigates how firefighting resources’ arriving time affects the fire suppression result. In these simulations, we make the fire suppression start from a fixed location and add variances to firefighting resources’ arriving time. Specifically, the initial dispatch location is fixed to cell (102, 101), which is at the tail area of the fire. The arriving time of each resource is sampled from $normal(\mu_s, \sigma^2)$, where μ_s is the scheduled arriving time of the resource as given in Table IV, and $\sigma^2 = 600s$ (10 minutes). Twenty simulations were run, and Table VI shows the average fire suppression time, finish time, fire perimeter, burned area, and standard deviation of fire perimeter and burned area of these 20 simulations. The results show that an arrival time variance of 10 minutes in this case does not have a significant impact on the final suppression results. This can probably be attributed to the randomly generated arrival times, which make some resources arrive early and some later, thus complementing each other.

Finally, to investigate how the production rate of firefighting resources affects fire suppression result, we add variances to the agent’s production rate.

Table VI. Results of Fire Suppression for Different Deployment Times

Dispatch Time	Suppression Time (hrs)	Finish Time (hrs)	Fire Perim. (km)	stdev	Burned Area (ha)	stdev
$\mu_s = \text{scheduled}$ ($\sigma^2 = 600s$)	2.60	3.10	2.05	0.100	26.46	2.23

Table VII. Results of Fire Suppression for Different Production Rates

Prod. Rate ($\sigma^2 = 0.10\mu$)	Suppression Time (hrs)	Finish Time (hrs)	Fire Perim. (km)	stdev	Burned Area (ha)	stdev
$\mu = 0.90\mu_s$	2.82	3.36	2.20	0.055	30.52	1.30
$\mu = 0.95\mu_s$	2.71	3.21	2.12	0.054	28.36	1.35
$\mu = 1.00\mu_s$	2.60	3.10	2.05	0.040	26.63	1.11
$\mu = 1.05\mu_s$	2.50	3.00	1.98	0.041	24.84	0.87
$\mu = 1.10\mu_s$	2.36	2.86	1.90	0.041	22.94	0.85

Here the arriving time is as scheduled and the initial dispatch location is fixed to (102, 101). After every step that the agent finishes suppressing a cell, the agent's new production rate is sampled from a $normal(\mu, \sigma^2)$. Let μ_s denote the agent's standard production rate as given in Table II and ρ be some factor. Then we calculated the value for μ as $\mu = \rho\mu_s$, where ρ was varied from 0.90, 0.95, 1.00, 1.05, and 1.10. The variance was arbitrarily set to $\sigma^2 = 0.10\mu$ for demonstration purpose. We note that in real firefighting, the fireline production rate and variance vary considerably for different fuels, topography, and resource types. Some information can be found in Lee et al. [1991]. With these parameters, we study how the decrease and increase of production rates may affect the suppression results. We made 20 simulation runs for each value of ρ . Table VII shows the average fire suppression time, finish time, fire perimeter, burned area, and the standard deviation of fire perimeter and burned area of these simulations. The results show that for the same mean production rate μ , the 10% variance does not make much difference in the suppression results (as indicated by the small values of σ). This is because the random generated production rates in different steps averaged out over the long run. Table VII shows that as the mean production rate μ increases, the fire suppression time, fire perimeter, and burned area decrease. Specifically, as μ increases from $1\mu_s$ to $1.1\mu_s$, the suppression time decreases from 2.60 hours to 2.36 hours; as μ decreases from $1\mu_s$ to $0.9\mu_s$, the suppression time increases from 2.60 hours to 2.82 hours.

This experiment illustrates how fire spread simulation, firefighting resource optimization, and fire suppression simulation can effectively work together to provide valuable information for fire containment. An important result revealed in this experiment is that when firefighting tactics are considered, the firefighting resources suggested by the optimization component can actually suppress the fire earlier than calculated by the optimization algorithm, and result in much less fire perimeter and burned area. For example, based on the optimization algorithm, the fire is suppressed at around 4 hours with perimeter around 3.49km. The data in Table V shows that the average finish time (of all four dispatch zones) is 3.12 hours and average fire perimeter is 2.11km. This difference

is because the interaction between fire suppression and fire spread slows down the fire spread, and thus changes the “free burn” fire spreading assumption made by the optimization algorithm. As a result, the dispatch plan computed by the optimization algorithm “overestimates” the firefighting resources that are needed. One can view the optimization model as providing an upper bound for the needed resources that guarantee containing the fire within the computed time. Nevertheless, by employing an iterative simulation-optimization scheme between the optimization model and fire suppression simulation, the decisions computed by the optimization model can be refined. This point is discussed further in Section 6.

5.3 Results with Real GIS Data

The above two experiments use simplified GIS settings in order to better show the effect of different firefighting tactics. This section extends the previous experiments to demonstrate the integrated framework using real GIS data. With real GIS data, different cells can have different fuel models, slopes, and aspects. Together with changing wind speed and direction, they give irregular fire spreading shapes. Despite this difference, the proposed simulation and optimization framework follows the same procedure as described above.

In this experiment, the same GIS data defined in the Ashley project in FAR-SITE was used by DEVS-FIRE to generate fire spreading scenarios. The wind speed was randomly generated from $uniform(4, 5)$ miles/hour and the wind direction was sampled from $uniform(-30, 60)$ every half an hour. Fire was ignited at cell (180,160). We note that these data points are arbitrarily selected for experiment purpose, which may not represent reality well. Time-indexed fire perimeters and burned areas were recorded for each scenario. To compute the firefighting resource dispatch plan, the same resources given in Table II were used. Then fire suppression simulations were set up where all resources work as one group to build fireline in clockwise direction. Due to space limitation, detailed results of fire spread simulations, optimizations, and fire suppression simulations are omitted here. Some pictures of the fire spread simulations using the Ashley GIS data can be found in Gu et al. [2008]. In the following we take a sample scenario and highlight some major results. In this scenario, the hourly fire growth perimeters and burned areas are (0.57, 3.15), (1.28, 11.61), (2.44, 33.75), (3.73, 79.72), (5.08, 140.85), (6.82, 221.58) (km, ha) for the 6 hours. The dispatch plan computed by the optimization algorithm suggests that all the seven resources in Table II should be deployed and the fire is suppressed at the end of 4 hours. With these resources, fire suppression simulations (using the one-agent direct attack configuration) show that the average fire suppression finish time is around 3 hours, with finished perimeter around 2.1km. Specific finish time and fire perimeters vary for different dispatch locations. From these results one can make similar conclusions as before.

Two things are worth pointing out when using real GIS data. First, the calculation of the fire perimeter needs to take into account the slope factor. Second, in real firefighting the production rate of a firefighting resource is affected by the fuel model and the slope. Discussion on how firefighting resources’ production

rates depend on fuel model and slope can be found in the fireline handbook [NWCG 2004] and [Caballero 2002]. This experiment has not taken into account such realistic factors of using different production rates for different fuel models. However, since the agent constructs the fireline in a stepwise fashion from one cell to another, these factors can be incorporated into the calculations in future work.

6. DISCUSSION

Decision making for wildfire containment generally requires input from various sources, including prediction of fire behavior, optimization of firefighting resources, and simulation of firefighting tactics. Even though extensive research has been conducted in each of these areas in the literature, without an integrated framework the dependencies and interactions among these components cannot be studied in an effective manner. This article represents efforts towards the development of such an integrated framework for supporting decision making by fire managers. Within this framework, fire managers can generate dispatch plans of firefighting resources based on the simulated fire spread behavior and stochastic optimization. Such plans are then quantitatively evaluated by simulation of firefighting tactics, which in turn provide information that could further refine the dispatch plans. Simulation of firefighting tactics also allows for comparing different firefighting configurations to provide valuable information for implementing a dispatch plan in the field. This work builds a foundation where more advanced optimization algorithms and simulation configurations can be incorporated in the future. Next we discuss the limitations of the current work and provide directions for future work.

The three experiments in Section 5 illustrate the agent-based fire suppression simulation from different aspects. Due to space limitation, this article mainly presents the results based on direct attack. It is expected that results of parallel attack will follow similar patterns but with larger perimeters and burned areas. The result of an indirect attack largely depends on the firefighting route specified by the fire manager, which can utilize landscape characteristics such as road and river. We note that all the experiments in this paper are based on somehow simplified, and maybe uncommon, firefighting configurations when compared to real firefighting. For example, it is said that fire attack in single agent mode is rarely used, and direct head attack is usually not applicable for fires that matter (that spread fast and have greatest potential to become big and damaging). Furthermore, real firefighting will most likely involve more dynamical configurations, for example, combining all the three attack tactics according to specific situations.

The agent-based fire suppression model adopts a grid-based representation because the underlying fire spread model is a cellular space model. This grid-based approach brings advantages such as easier incorporation of different fuels/topography and agent starting points. On the other hand, because of the grid-based representation, an agent can only proceed from center to center of cells, thus introduces approximation errors when compared with a vector-based approach such as the one developed in Fried and Fried [1996]. In particular,

because an agent always selects a cell that is “outside” the fire perimeter in order to “safely” enclose the fire, the effect of this error on fire perimeter and burned area will always be biased, i.e., producing larger perimeter and burned area. An important part of future work is to quantitatively evaluate the magnitude of this error by comparing with a vector-based approach such as FCAT (Fire Containment Algorithms Tester) [Fried 1996] or FARSITE [Finney 1998]. Systematic comparison methods and metrics will be developed to characterize the error from different aspects such as grid resolution, size of fire, and different attack tactics.

Integrating simulation and optimization for effective fire containment is one of the major goals of this work. The focus of this article has been to demonstrate how to carry out fire spread simulation, firefighting resource optimization, and then fire suppression simulation in an integrated manner. However, the feedback loop from fire suppression simulation to optimization has not been dealt with in detail. The weakness of the optimization model WFCP (1-2) could be that this model does not account for the interaction of the fire-front with the constructed fireline. Consequently, the model will generally tend to overestimate the firefighting resources that are needed as shown in our experimental results. Therefore, an iterative procedure via simulation-optimization [Fu 1994] between the optimization model and the fire suppression simulation can be implemented to refine the dispatch plans generated by the optimization component. Next we briefly outline one such iterative procedure.

Let k denote the iteration index and let $x^k = \{x_j^k\}_{j \in J}$ be the first-stage optimal solution to WFCP specifying the resources to dispatch to the fire(s). Let J_0^k and J_1^k denote the sets of indices $j \in J$ such that $x_j^k = 0$ and $x_j^k = 1$, respectively. Also let $n_1^k = |J_1^k|$, where $|\cdot|$ denotes the cardinality of the set. As pointed out in Section 2, three cases are possible after running the fire suppression simulation using the dispatch plan defined by x^k : Case (1) x^k is “satisfactory” to contain the fire(s); Case (2) x^k overestimates the resources that are needed to contain the fire(s); and Case (3) x^k underestimates the resources that are needed to contain the fire(s). We propose an iterative scheme that involves appending linear constraints to the first-stage formulation (1) to cut off x^k if this solution falls under cases 2 and 3. In Case 2 this can be achieved by appending the constraint $\sum_{j \in J_1^k} x_j \leq n_1^k - 1$ to the first-stage (1), and then reoptimizing WFCP to get a new solution x^{k+1} to pass to the fire suppression simulation for evaluation. Similarly, under Case 3 the following constraint can be added: $\sum_{j \in J_0^k} x_j + \sum_{j \in J_1^k} x_j \geq n_1^k + 1$. Note that to maintain feasibility of WFCP an artificial variable, restricted to be nonnegative, can be added to either constraint and penalized in the objective function. This can be done to accommodate the case when there is no dispatch plan that can actually contain the fire. Future work will study such an iterative procedure considering various rules for defining each of the three cases as well as convergence of the method. Instead of using WFCP, we will also consider using scenario-based standard response models [Haight and Fried 2007], which do not explicitly consider the interaction of the firefront and the constructed fireline.

7. CONCLUSION

Wildfires pose a serious threat to communities and ecosystems throughout the world. The ability to predict fire spread behavior, to optimize a plan for firefighting resources management, and to evaluate the plan before dispatching resources to the field is essential for supporting decision-making concerning fire containment. This article presents an integrated framework and demonstrates how fire spread simulation, firefighting resource optimization, and fire suppression simulation can effectively work together for wildfire containment. Within this framework, the article focuses on the aspect of fire suppression simulation and presents comprehensive results for different firefighting tactics. The experimental results show that different firefighting tactics, including different types of attack, initial deployment locations, and group configurations, can lead to significantly different fire suppression results. Thus it is important to take into account these firefighting tactics when making a decision for suppressing a fire. When integrated with optimization of firefighting resources, simulations of these tactics with stochastic variables can provide valuable information for fire managers to examine the details of a dispatch plan before actual implementation in the field. Future work includes adding more realistic factors such as different resource production rates for different fuel models/slopes, considering multiple wildfires, characterizing the approximation error introduced by the grid-based approach, validation of the integrated approach using real wildfires, and extending the stochastic optimization model to take into account the interaction between fireline construction and wildfire spread.

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